

- (b) If $\alpha < \alpha^*$, a stricter performance target has an ambiguous effect on the virality of misinformation;
- (c) If $\alpha > \alpha^*$, a stricter performance target reduces the virality of misinformation even when the platform optimally chooses the sharing network conditional on the policy.

This result captures similar ideas to those in Propositions 5 and 6. When performance targets are well-enforced through auditing, reducing the target (making it stricter) necessarily curbs the spread of misinformation. When the platform is held more accountable for misinformation shares, it reduces the misinformation in circulation, even if it happens to be more sensational. Lower targets therefore align regulator and platform incentives to remove the least reliable content.

However, with a lower auditing probability, the incentives of the regulator and platform diverge. When performance targets are lax, the platform still has some incentive to comply: it can remove low-reliability, low-sensational content and meet the performance standard without much of a hit to engagement. However, as the performance target continues to tighten, the platform needs to remove low-reliability sensationalist content, which would significantly reduce engagement. In this case, relying on the moderate probability of auditing, the platform may prefer to violate the performance target and district targets backfire.

6.4 Network Regulations

As we saw in Theorem 3, when unregulated, the platform (endogenously) chooses the island model of Section 4 with parameters (p_s, p_d) . Here, we consider limits on the ideological homophily induced by the platform's algorithm. In particular, we suppose the regulator can choose a homophily standard, based on the ratio between within-island links to across-island links.

Proposition 8. *There exists $\gamma < \infty$ such that for any $p^* \geq \gamma$, if the regulator imposes $p_s/p_d \leq p^*$ then (i) the platform chooses the island model with $p_s/p_d \leq p^*$; and (ii) the virality of misinformation is reduced.*

The regulator can thus achieve its objective by moderating the homophily that the platform can impose on the sharing network. In particular, algorithmic choices that create extreme homophily (as demonstrated in Theorem 3(a)), can be curbed by imposing a standard that requires content to be shared across ideological lines. This is closely related to the “ideological segregation standard” discussed in Sunstein (2018), which aims to restrict the extent to which content is curated specifically to the ideology or interests of the group of users. Such standards ensure that echo chambers are broken and users of differing ideology interact more frequently, limiting the spread of misinformation.

We note that the regulation in Proposition 8 is not always binding for the platform. In the case of highly reliable articles (as in Theorem 3(b)), the platform maximizes engagement by implementing a maximally-connected sharing network, so the homophily constraint is moot. However, when the article is less reliable, the regulation will bind and the platform will be forced to maintain a minimum level of connectivity.